with a full 2×2: each conditioning architecture (pooled H2, full-sequence H4) trained with and without the interaction pipeline, all evaluated under the unified pipeline.

Table 7: Training-data pipeline 2×2 (FID ↓, unified eval, 3 replications; best checkpoint per cell). The two interventions are approximately additive: conditioning granularity dominates (≈2.7–2.8 FID in both columns), and interaction-aware sampling contributes a similar absolute gain to both architectures (−0.59 pooled, −0.51 full-sequence).

	− interaction sampling	+ interaction sampling
H2 (pooled)	6.214 ±0.105	5.620 ±0.117
H4 (full-sequence)	3.392 ±0.179	**2.880** ±0.084

Two conclusions follow. First, the conditioning-granularity gap (≈2.7–2.8 FID in both columns) is roughly five times the pipeline effect, confirming that full-sequence conditioning — not data curation — is the primary driver. Second, the two interventions are approximately additive: interaction-aware sampling contributes a similar absolute gain to both architectures (−0.59 pooled, −0.51 full-sequence), and for the full-sequence model the best checkpoint additionally arrives at $1/3$ the epochs. EgoPed-IA, the +pipeline full-sequence cell, is our best system.

5.3 Training-Seed Sensitivity

We retrain H4 from two additional seeds under its original recipe and select each seed's best checkpoint by training-time validation FID (the same protocol as the main results). Definitive best-checkpoint FID across the three seeds is 3.392, 4.719, and 4.803 (mean 4.30 ± 0.79): the headline seed is markedly better than the other two, so single-seed point estimates in this setting overstate precision, and we report the spread explicitly. The cross-*architecture* conclusions are unaffected — every H4 seed remains well below the pooled baseline (5.620), and the H2 baseline itself proved training-unstable (two additional H2 seeds remained above FID 10.9 within twice the original training budget), meaning all our comparisons use the baseline's *best* seed and are conservative in the baseline's favor. EgoPed-IA is reported as a single seed.

5.4 Encoder Freeze vs. Unfreeze (H6)

The most important ablation is whether to freeze the ego encoder during diffusion training. H6 uses an identical architecture to H4 but with FREEZE_EGO=False, allowing the encoder to co-adapt with the denoiser via backpropagation through the diffusion loss.

Figure **??** shows the H6 training trajectory (FID and R@1 vs. epoch). Two behaviors are notable:

- **FID/R@1 tradeoff**: As training progresses, R@1 improves (encoder specializes discriminatively) but FID degrades (generation quality decreases). The two optima do *not* coincide: training-time FID is best at epoch 2099 (4.789), while R@1 continues rising long after FID has begun to deteriorate.
- **R@1 saturation far below H4**: After epoch ≈3299, training-time R@1 saturates, oscillating in the 0.37–0.42 band through epoch 4999 (maximum 0.417 at epoch 4599) while FID remains between 5.1 and 6.3. Neither metric ever approaches H4's operating point (FID 3.392, R@1 0.548) at any point in 5000 epochs.

Definitive evaluations (3 replications, CFG = 10) confirm the picture:

H6 is worse than H4 on *all metrics at all evaluated checkpoints*. Later checkpoints cannot change this conclusion: even H6's best training-time R@1 in the saturation band (0.417 at epoch 4599, with training-time FID 6.339) remains far below H4's definitive R@1 of 0.548 — and training-time metrics consistently *overestimate* the definitive 3-replication values (e.g., 0.398 → 0.348 at epoch 3299). The MultiModality collapse is particularly striking: H6 MM = 2.618 vs. H4 MM = 3.956 (34% lower). This indicates near-deterministic generation per condition — the ego encoder has learned highly discriminative features for each trajectory, collapsing the generated motion distribution to a narrow region.

Table 8: H6 definitive evaluation at three checkpoints. H4 reference included for comparison.

System	FID ↓	R@1 ↑	MM ↑	Diversity
H6 epoch=2099 (best train FID)	4.952 ±0.088	0.263 ±0.009	2.455	5.957
H6 epoch=3299 (best seg-1 R@1)	5.311 ±0.060	0.348 ±0.004	2.581	6.034
H6 epoch=3399 (matched to H4 best)	5.079 ±0.034	0.352 ±0.009	2.618	6.064
H4 epoch=3399 (FREEZE=True)	**3.392** ±0.179	**0.548** ±0.002	**3.956**	5.794

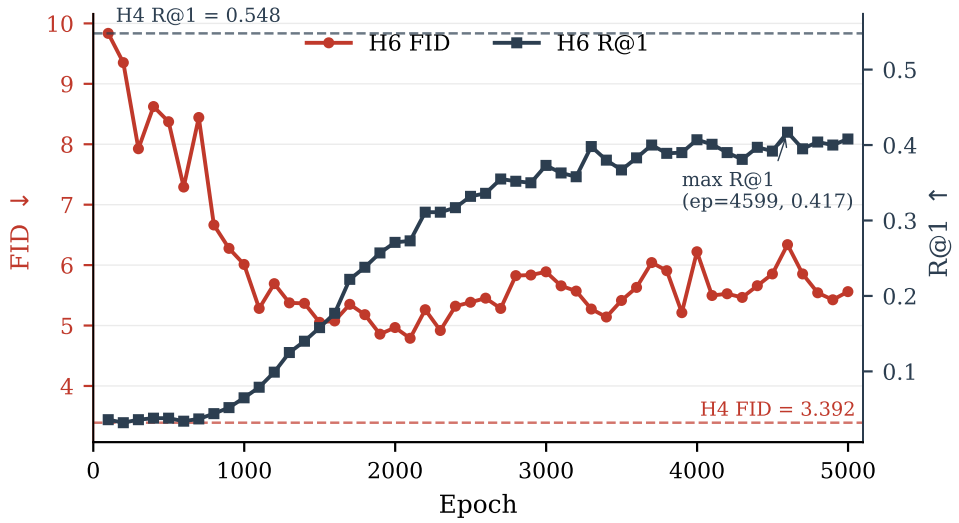

Figure 3: H6 training trajectory (FREEZE_EGO=False): training-time validation FID (left axis, red) and R@1 (right axis, blue) vs. epoch, over the full 5000-epoch run. FID is best at epoch 2099 and degrades as R@1 rises; after epoch ≈3299, R@1 saturates in the 0.37–0.42 band while FID oscillates between 5.1 and 6.3. Dashed reference lines show H4's best operating point (FID=3.392, R@1=0.548 at epoch=3399, CFG=10). H6 never approaches H4's performance on either metric.

Mechanism. When the encoder is frozen (H4), it maps ego trajectories to general-purpose representations aligned with the VAE latent space. The denoiser learns to query these fixed representations and generate diverse, plausible motions for each condition. When the encoder is unfrozen (H6), backpropagation through the denoising loss pushes the encoder toward representations that *minimize the denoising loss* — essentially learning discriminative features that uniquely identify each trajectory. This over-specialization constrains generation: the denoiser outputs similar samples for each unique ego representation, collapsing MultiModality.

This pattern mirrors the frozen-encoder paradigm in image generation **???**: frozen conditioning encoders preserve general-purpose representations and allow the generative model to learn diverse ways to satisfy the conditioning signal.

Is the pretraining necessary? Freezing alone is not sufficient: an H4 model trained with a *randomly initialized* frozen ego encoder (identical otherwise) reaches only FID $= 4.968 \pm 0.171$ with R-Precision@1 exactly at chance — statistically indistinguishable from the model's unconditional prior (FID 5.18, Table **??**). Together with H6, this brackets the design from both sides: the contrastive pretraining is what *injects* usable ego information, and freezing is what *preserves* generative diversity while using it.

5.5 Conditioning Mechanism: Self-Attention vs. Cross-Attention

Our denoiser injects the ego sequence by concatenating it with the latent tokens and applying self-attention over the joint sequence (Section **??**). A common alternative for sequence conditioning is *cross-attention*, where the latent tokens act as queries and the ego sequence as keys and values, so the ego tokens never attend among themselves and the attention cost is $\mathcal{O}(L \times T)$ rather than

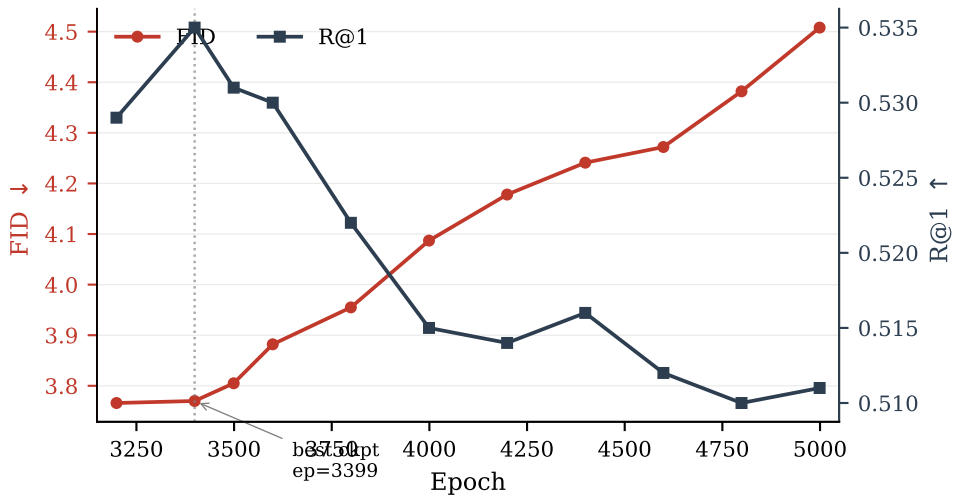

Figure 4: H4 training trajectory: training-time validation FID (left axis, red) and R@1 (right axis, blue) across the late training segment. Epoch 3399 is the best checkpoint (highest R@1, FID within noise of epoch 3199, and best definitive 3-replication FID); both metrics then degrade steadily through epoch 4999. Training loss is flat across this range — sampling quality oscillates independently of training loss.

$\mathcal{O}((L+T)^2)$. To test whether the specific mechanism matters, we train an otherwise-identical cross-attention denoiser (same frozen VAE and frozen ego encoder, same data, same effective batch size) and evaluate it on the held-out test split.

Table 9: Conditioning-mechanism ablation. Both denoisers consume the full 196-token ego sequence and differ only in how they attend to it. Evaluated on the held-out test split (a scene-disjoint half of the validation set), CFG = 10, mean ± 95% CI over 3 replications, best checkpoint each. The two mechanisms are statistically comparable (overlapping FID intervals); cross-attention provides no advantage.

Denoiser	Attention over ego	FID ↓	R@1 ↑	MM ↑
Self-attention (EgoPed)	concatenation	**3.338** ±0.190	**0.537**	4.14
Cross-attention	queries/keys-values	3.490 ±0.059	0.501	**4.61**

As shown in Table **??**, the self-attention and cross-attention denoisers achieve statistically comparable FID (3.338 vs. 3.490; 95% CIs $[3.15, 3.53]$ and $[3.43, 3.55]$ overlap), with self-attention holding a slight edge on retrieval alignment (R@1) and cross-attention generating marginally more diverse motion per condition (MM). Note that the comparison is, if anything, generous to the cross-attention variant: its best checkpoint was selected on the held-out split itself, whereas the self-attention model's checkpoint was fixed in advance from training-time validation. We also observed that the cross-attention variant trained less stably, with a noisier FID-vs-epoch trajectory. **Takeaway**: the large improvement over pooled conditioning (Table **??**) comes from exposing the denoiser to the *full* per-timestep ego sequence, not from any particular attention mechanism – either self-attention over the concatenated sequence or cross-attention works, and neither is required over the other.

5.6 Checkpoint Selection

A non-obvious finding (relevant to practitioners) is that FID does not monotonically improve with training. H2's epoch trajectory shows FID $= 6.603$ at epoch 4399, FID $= 8.400$ at epoch 4599 (regression), and FID $= 7.510$ at epoch 4999 (partial recovery) — despite flat training loss throughout. H4 similarly peaks at epoch 3399 (definitive FID $= 3.392$) and degrades steadily thereafter (Fig. **??**). **Lesson**: the final checkpoint is not the best checkpoint. Periodic FID validation during training is essential; early stopping on FID is critical.

6 Discussion

Architecture recommendation. The experimental evidence strongly favors the H4 architecture: frozen ego encoder + full-sequence conditioning denoiser (self-attention over the concatenated latent and ego sequence; cross-attention works equally well, Section **??**). Future work seeking to improve conditioning fidelity (R@1) should explore adapter-style approaches — fine-tuning a lightweight adapter on the frozen encoder while keeping the backbone fixed — rather than end-to-end unfreezing.

R-Precision@1 and the conditioning-diversity tradeoff. H4's R@1 (0.548) is lower than H2's (0.671) despite dramatically better FID. This seemingly paradoxical result reflects a genuine tradeoff: richer conditioning via the full ego sequence enables the denoiser to generate *more diverse* motions per condition, spreading the generated embedding distribution and reducing retrieval accuracy. In the context of AV simulation, diversity within valid behaviors is a feature — a single ego trajectory should plausibly lead to multiple pedestrian behaviors (crossing, waiting, accelerating). R@1 measures only conditioning alignment, not generation quality; FID captures the latter more faithfully.

Limitations.

- **Body pose quality**: Pedestrian poses are estimated (not ground-truth 3D), introducing noise in training labels. Direct 3D body annotation from datasets like JRDB-Pose or BEDLAM would improve quality.
- **Conditioning scope**: We condition only on 2D ego trajectory. Incorporating relative pedestrian-ego geometry, ego velocity profile, or map elements could improve fidelity.
- **R@1 gap**: H4 achieves R@1 $= 0.548$, below H2's 0.671. The adapter-style approach described above is a natural next step.
- **Physics**: Generated motions are not physically constrained (foot sliding, penetration). Post-hoc physics cleanup could improve realism for simulation deployment.
- **Evaluator domain shift**: FID and Diversity rely on motion-feature encoders pretrained on HumanML3D, whose motion distribution differs from in-the-wild pedestrian motion; absolute FID values should be compared only within our benchmark.
- **Single benchmark, no perceptual study**: All results are on our combined AVA/nuScenes/Waymo benchmark, and we do not yet report a human perceptual study of motion realism.

Broader impact. The primary intended use of ego-conditioned pedestrian motion generation is autonomous-driving simulation: synthesizing realistic, ego-aware pedestrian behaviors enables larger-scale safety testing without exposing real pedestrians to risk, and is particularly valuable for the long tail of rare or safety-critical interactions that are unsafe or impractical to collect. We note two risks. First, generated pedestrian motions could in principle be used adversarially, e.g. to seed motion-planning policies that learn to expect only nominal behaviors and consequently fail on out-of-distribution pedestrians; we mitigate this by reporting MultiModality and diversity metrics, which expose the breadth of the generated distribution rather than only its mean. Second, training data inherit demographic biases of the source AV datasets (AVA, nuScenes, Waymo): generated motion is conditioned on the distribution of pedestrians actually observed in U.S./European urban environments, so deployment in other settings should be preceded by additional validation. Generated motion data is synthetic and contains no personally identifiable information.

7 Conclusion

We introduced EgoPed, an ego-conditioned pedestrian body motion generation system based on latent diffusion with full-sequence ego conditioning. Our key finding is that exposing the denoiser to the full 196-token ego trajectory sequence dramatically outperforms single-token pooled conditioning: a 40% FID improvement (5.620 $\rightarrow$ 3.392) under a unified evaluation pipeline, with non-overlapping confidence intervals, verified on a scene-disjoint held-out split, on evaluator-independent trajectory metrics, and conservative with respect to both training-data curation and baseline seed selection. Interaction-aware data sampling amplifies the result to FID = 2.880. A paired ablation reveals that

frozen ego encoders are critical: unfreezing causes 34% MultiModality collapse as the encoder over-specializes to discriminative features.

These results establish a clear design recipe for ego-conditioned motion generation in AV settings: train the ego encoder contrastively on frozen VAE latents, then freeze it during diffusion training and condition the denoiser on the full ego sequence. This recipe produces state-of-the-art ego-conditioned full-body pedestrian motions across AVA, nuScenes, and Waymo, and provides a foundation for scalable, behavior-consistent pedestrian simulation.

A Implementation Details

Ego encoder. 4-layer transformer encoder, 4 attention heads, 256D hidden dimension (1024D feed-forward), sinusoidal positional encoding for $T = 196$ timesteps. Input: 2D ego position sequence, projected linearly to 256D. No projection head (we verified the 2-layer MLP projection head does not improve val MSE: 1.919 vs. 1.920). Pretraining: AdamW (lr $= 1 \times 10^{-4}$, weight decay 10^{-4}), 200 epochs, batch size 64, linear warmup followed by cosine annealing, gradient clipping at max-norm 1.0.

Diffusion model. MLD denoiser, `trans_enc` architecture (self-attention over the concatenated latent, timestep, and ego tokens): 9 layers, 4 attention heads, 256D token dimension, 1024D feed-forward, latent sequence length $L = 4$. The cross-attention variant of Section **??** uses the same hyperparameters in a `trans_dec` configuration (latent tokens as queries, ego sequence as keys/values). Diffusion: 1000 DDPM training steps (scaled-linear β schedule), DDIM inference (50 steps, $\eta = 0$). CFG dropout rate: 10%. Optimizer: AdamW, lr $= 1 \times 10^{-4}$ (constant), 5000 total epochs. Batch size: 128. Training hardware: NVIDIA H100 (80 GB) on an academic HPC cluster; training runs in segments of up to 24h wall time.

Checkpoint selection. We select checkpoints based on validation FID (not training loss). For H4, epoch 3399 is best; we confirm this with the full training trajectory and definitive 3-replication evaluation. For H2, epoch 4399 is best despite continued training to epoch 5000.

Evaluation. All evaluations use `TEST.REPLICATION_TIMES=3` (3 independent generation passes), `DIVERSITY_TIMES=300`, `MM_NUM_SAMPLES=100`, `MM_NUM_TIMES=10`. Each replication regenerates the full test set with fresh sampling noise; the diffusion sampler is stochastic across replications while the data pipeline and evaluator are fixed (training uses seed 1234). Confidence intervals are reported as $\pm$ CI (half-width of the 95% interval across replications).

B H6 Training Trajectory Table

C NeurIPS Paper Checklist

1. **Claims.** *Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?*
   Answer: [Yes]. Justification: The headline claims (40% FID improvement from full-sequence conditioning under a unified evaluation pipeline; further gain from interaction-aware sampling; MultiModality collapse from unfreezing the ego encoder) are supported by replicated evaluations with confidence intervals in Tables **??**, **??**, and **??**, seed-sensitivity reporting in Section **??**, and held-out/evaluator-independent verification in Sections **??** and **??**.

2. **Limitations.** *Does the paper discuss the limitations of the work?*
   Answer: [Yes]. Justification: Section **??** lists estimated (not ground-truth) pose labels, restricted conditioning scope, the R@1 gap, lack of physical constraints, evaluator domain shift, and the absence of a perceptual study.

3. **Theory.** *Does the paper include theoretical results?*
   Answer: [NA]. Justification: This is an empirical systems paper; no theoretical claims are made.

Table 10: H6 (FREEZE_EGO=False) training-time validation metrics at selected checkpoint evaluation points (validation runs every 100 epochs; full curve in Figure **??**). Training-time FID is best at epoch 2099 (4.789); R@1 rises until epoch ≈3299, then saturates in the 0.37–0.42 band (maximum 0.417 at epoch 4599) while FID remains between 5.1 and 6.3.

Epoch	Train-time FID ↓	Train-time R@1 ↑
99	9.835	0.036
499	8.373	0.038
999	6.011	0.065
1499	5.053	0.158
1999	4.967	0.271
2099	**4.789**	0.273
2499	5.385	0.332
2999	5.889	0.373
3299	5.273	0.398
3399	5.141	0.380
3499	5.415	0.367
3999	6.221	0.407
4599	6.339	**0.417**
4999	5.561	0.408

4. **Experimental reproducibility.** *Does the paper fully disclose the information needed to reproduce the main experimental results?*
   Answer: [Yes]. Justification: Architecture, training, and evaluation hyperparameters are given in Section **??** and Appendix **??**; the dataset construction pipeline is described in Section 4.1.
5. **Code and data access.** *Does the paper provide open access to data and code?*
   Answer: [No]. Justification: Code and the derived dataset will be released upon publication; the underlying nuScenes and Waymo data are publicly available under their research licenses, and the AVA recordings will be released subject to anonymization requirements.
6. **Experimental settings.** *Does the paper specify all training and test details?*
   Answer: [Yes]. Justification: Optimizer, learning rate, batch size, epochs, CFG dropout, sampler settings, and checkpoint-selection protocol are in Appendix **??**; evaluation settings (replications, metric parameters) in Section **??** and Appendix **??**.
7. **Error bars.** *Does the paper report error bars or other statistical significance information?*
   Answer: [Yes]. Justification: All key metrics report 95% confidence intervals over 3 independent generation replications; the headline FID comparison has non-overlapping intervals.
8. **Compute resources.** *Does the paper provide sufficient information on the compute resources?*
   Answer: [Yes]. Justification: Single NVIDIA H100 (80 GB) per run, $\leq$24h wall-time segments, $\sim$2 segments per diffusion model (Appendix **??**); evaluations take $\sim$8 minutes each.
9. **Code of ethics.** *Does the research conform to the NeurIPS Code of Ethics?*
   Answer: [Yes]. Justification: The work uses licensed public AV datasets and our own staged recordings with consenting participants; generated data contains no personally identifiable information.
10. **Broader impacts.** *Does the paper discuss potential positive and negative societal impacts?*
    Answer: [Yes]. Justification: Section **??** discusses safety benefits for AV simulation, adversarial-use and dataset-bias risks, and mitigations.
11. **Safeguards.** *Does the paper describe safeguards for responsible release?*
    Answer: [NA]. Justification: The model generates pedestrian body motion for simulation and poses no high misuse risk requiring gated release.
12. **Licenses.** *Are existing assets properly credited and licenses respected?*
    Answer: [Yes]. Justification: nuScenes and Waymo are used under their research licenses

and cited; the MLD backbone, HumanML3D representation and evaluators, OmniRe, MS3D, and SMPL are cited.

13. **New assets.** *Are new assets well documented?*
    Answer: [Yes]. Justification: The dataset construction pipeline (annotation, pose extraction, interaction cropping) is documented in Section 4.1; documentation will accompany the release.

14. **Human subjects.** *Does the paper involve crowdsourcing or research with human subjects?*
    Answer: [Yes]. Justification: Staged interaction scenarios were recorded with informed, consenting participants; recordings follow institutional data-protection procedures, and released data will be anonymized.